

GOSFORD HIGH SCHOOL



2016

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

- **General Instructions**
- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A Reference Sheet is provided at the back of this paper
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total Marks – 70

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II Pages 6 – 12

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the value of $\lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$

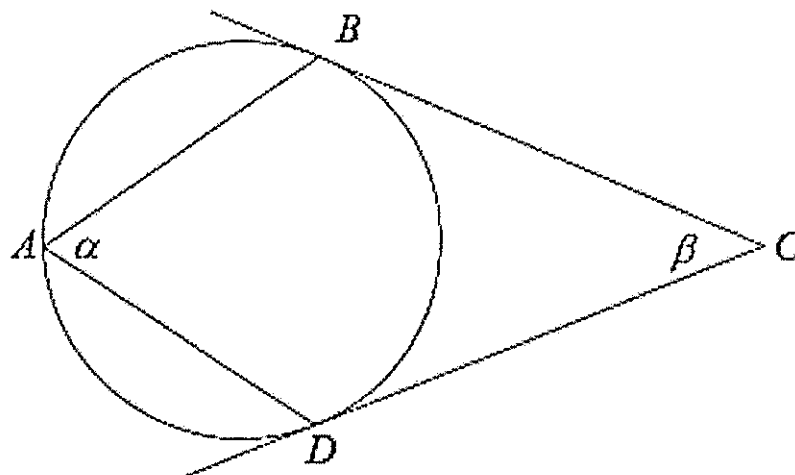
(A) 0

(B) $\frac{2}{3}$

(C) 1

(D) $\frac{3}{2}$

2. In the diagram below, BC and DC are tangents. Which statement is correct?



(A) $\alpha + \beta = 180^\circ$

(B) $2\alpha + \beta = 180^\circ$

(C) $\alpha + 2\beta = 180^\circ$

(D) $2\alpha - \beta = 180^\circ$

3. When the polynomial $P(x) = x^4 + ax + 2$ is divided by $x^2 + 1$ the remainder is $2x + 3$

The value of a is?

(A) 1

(B) 2

(C) 0

(D) 3

4. Which of the following is equal to $4^{\log_2 a}$?

(A) $2a$

(B) a^2

(C) a

(D) $4a$

5. Evaluate $\frac{\sin A}{\sin A + \cos A} - \frac{\sin A}{\sin A - \cos A} =$

(A) $\cot 2A$

(B) $\operatorname{cosec} 2A$

(C) $\tan 2A$

(D) $\sec 2A$

6. Which of the following is an expression for $\frac{d}{dx} \sin^{-1}(2x-1)$?

(A) $\frac{-1}{\sqrt{x(1-x)}}$

(B) $\frac{-1}{2\sqrt{x(1-x)}}$

(C) $\frac{1}{2\sqrt{x(1-x)}}$

(D) $\frac{1}{\sqrt{x(1-x)}}$

7. Find $\int \frac{1}{9+25x^2} dx$.

(A) $\frac{1}{15} \tan^{-1} \frac{5x}{3} + C$

(B) $\frac{1}{25} \tan^{-1} \frac{5x}{3} + C$

(C) $\frac{1}{25} \tan^{-1} \frac{3x}{5} + C$

(D) $\frac{1}{15} \tan^{-1} \frac{3x}{5} + C$

8. A particle is oscillating in Simple Harmonic Motion where its position x metres from a fixed point O on the same line as its motion after t seconds is given by $x = 2\cos\left(3t + \frac{\pi}{6}\right)$. What is the maximum speed of the particle?

- (A) 2 m/s
- (B) 6 m/s
- (C) 0 m/s
- (D) $\frac{\pi}{9}$ m/s

9. The solution to $|2x - 1| \leq |x - 2|$ is

- (A) $x \leq 1$
- (B) $x \geq 1$
- (C) $-1 \leq x \leq 1$
- (D) $x \leq -1$ or $x \geq 1$

10. A metal disc of 5 cm radius expands when heated. If the radius is increasing at a rate of 0.02 cm/sec, the rate at which the area of one of the faces is increasing is given by:

- (A) $\frac{\pi}{10}$ cm²/sec
- (B) $\frac{\pi}{5}$ cm²/sec
- (C) $\frac{2\pi}{5}$ cm²/sec
- (D) $\frac{5\pi}{2}$ cm²/sec

Section II

60 marks

Attempt Questions 11 – 14.

Allow about 1 hour and 45 minutes for this section.

Answer each question in a separate writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a new writing booklet.

- (a) Solve the inequality $\frac{3x-2}{x+1} > 2$ 2
- (b) $M(-1, 7)$ and $N(3, 1)$ are two points. If point $L\left(\frac{7}{2}, -\frac{5}{2}\right)$ divides the interval MN 2
externally in the ratio $k:1$, find the value of k . $\frac{1}{2}$
- (c) Find $\int \frac{1+6x}{1+x^2} dx$ 2
- (d) The two curves $y = x^3$ and $y = 2 - x^2$ intersect at $(1, 1)$. 3
Find the acute angle between the two curves at $(1, 1)$.
- (e) Use Mathematical Induction to show that for all positive integers n 3
 $1^2 + 3^2 + 5^2 + 7^2 + \dots + (2n-1)^2 = \frac{n}{3}(2n-1)(2n+1)$
- (f) Use the substitution $u = x^3 - 1$ to evaluate $\int_0^2 \frac{x^2}{(x^3-1)^2} dx$. 3

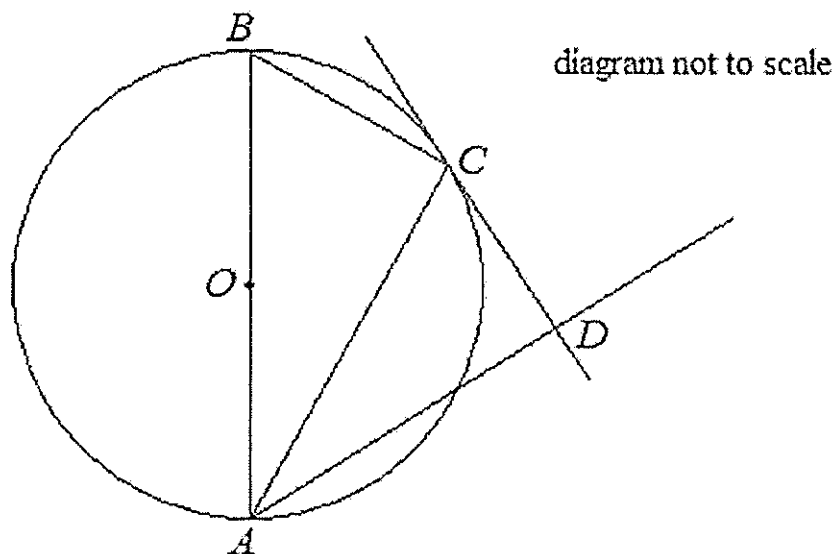
End of Question 11.

Question 12 (15 marks) Use a new writing booklet.

- (a) The three numbers a , b , c are consecutive terms in an arithmetic progression. 2

Show that the three numbers e^a , e^b , e^c are consecutive terms in a geometric progression.

- (b) 3



In the diagram AOB is the diameter of a circle centre O , and C is the point of contact of the tangent DC such that AC bisects $\angle DAB$.

Copy this diagram into your booklet.

Prove that AD is perpendicular to DC .

- (c) The polynomials $P(x)$ and $Q(x)$ are such that $P(x) = (x^2 - 1)Q(x) + ax + b$ for some constants a and b . $(x + 1)$ is a factor of $P(x)$ and when $P(x)$ is divided by $(x - 1)$ the remainder is 2. Find the remainder when $P(x)$ is divided by $(x^2 - 1)$. 3

Question 12 continues on page 8.

Question 12 continued

- (d) Find the exact area between the curve $y = \sin^{-1} x$, the x-axis and the lines $x = \frac{1}{2}$ and $x = 1$. **3**
- (e) Find the equation of the vertical and horizontal asymptotes of the curve $y = \frac{2x^2 + 1}{x^2 - 4x}$ **2**
- (f) For what values of x will $1 - \tan^2 x + \tan^4 x - \tan^6 x + \dots$ have a limiting sum for $0 \leq x \leq 2\pi$? **2**

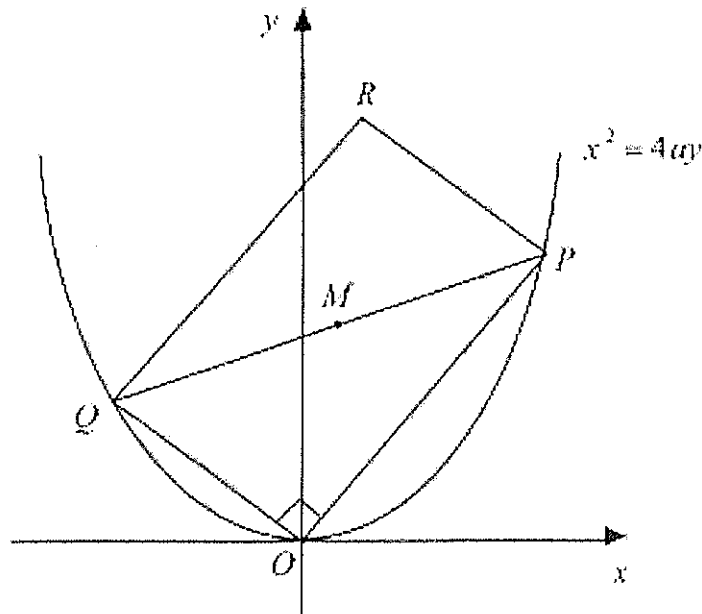
End of Question 12.

Question 13 (15 marks) Use a new writing booklet.

(a) Show that $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \sin^2 3x \, dx = \frac{1}{2} \left(\frac{\pi}{12} - \frac{1}{6} \right)$

3

(b)



$P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points which move on the parabola $x^2 = 4ay$ such that $\angle POQ = 90^\circ$, where O is the origin.

$M = \left(a(p+q), \frac{1}{2}a(p^2+q^2) \right)$ is the midpoint of PQ . R is the point such that $OPRQ$ is a rectangle.

(i) Show that $pq = -4$

1

(ii) Show that R has coordinates $(2a(p+q), a(p^2+q^2))$.

1

(iii) Find the equation of the locus of R .

2

Question 13 continues on page 10.

Question 13 continued

- (c) The acceleration of a particle is given by $\ddot{x} = 4(x + 1) \text{ ms}^{-2}$. Initially, the particle is at the origin and velocity is 2 ms^{-1} .

(i) Show that the velocity, v , at any position, x , is given by $v = 2x + 2$. 2

(ii) Hence show that $x = e^{2t} - 1$. 2

(d) (i) Show that $e^x + x = 3$ has a root between $x = 0$ and $x = 1$. 2

(ii) By taking $x = 0.8$ as an approximate solution, use one application of Newton's Method to find a better approximation, correct to 3 significant figures. 2

End of Question 13.

Question 14 (15 marks) Use a new writing booklet

- (a) The depth of water y metres in a tidal creek is given by $y = 5 - 4 \cos \frac{t}{2}$, for $0 \leq t \leq 4\pi$.

The time, t , being measured in hours.

- (i) Draw a neat sketch of $y = 5 - 4 \cos \frac{t}{2}$, showing all important features 2
- (ii) If the low tide one day is at 1.00 p.m., when is the earliest time that a ship requiring 3 m of water can enter the creek? Give your answer to the nearest minute. 2

- (b) At time t years the number, N of individuals in a population is given by $N = 500 - 400e^{-0.1t}$.

- (i) Show that $\frac{dN}{dt} = -0.1(N - 500)$. 1
- (ii) Find the population size for which the rate of growth of the population is half the initial rate of growth. 2

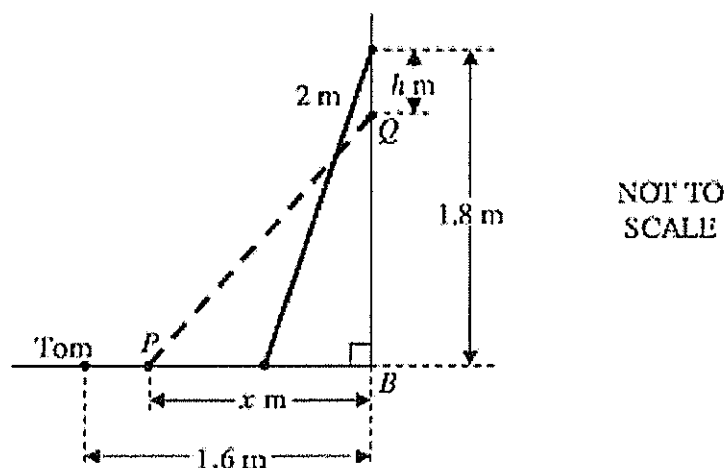
- (c) A group of 10 people, consisting of 6 girls and 4 boys, decided to go to the movies where they sit together in the same row.

- (i) How many ways can the 10 people be seated in a row? 1
- (ii) How many different arrangements of seating are possible where the 4 boys are all seated together? 2
- (iii) How many ways can at least one of the boys be separated from the other boys? 1

Question 14 continues on page 12.

Question 14 continued

- (d) The diagram shows a ladder PQ , 2 metres in length, leaning against a wall such that the top of the ladder, Q , initially reaches 1.8 metres up the wall. The base of the ladder, P , is x metres from the base of the wall, B .



The ladder begins to slide down the wall at the rate of 0.5 metres per minute such that the top of the ladder is h metres below its original position after t minutes.

- (i) Show that t minutes after the ladder begins to slide down the wall, 1

$$h = 1.8 - \sqrt{4 - x^2}.$$
- (ii) Tom is standing on the ground 1.6 metres from the base of the wall in a direct line with the ladder. 3
 At what rate does the base of the ladder hit Tom?

End of Exam.

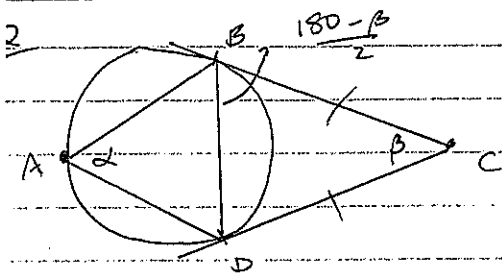
Extn 1

$$1 \quad \lim_{x \rightarrow 0} \frac{\sin 3x}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x \times \frac{2}{3}}$$

$$= \frac{3}{2} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x}$$

$$= \frac{3}{2} \quad (D)$$



$$\alpha = \frac{180 - \beta}{2} \quad (\angle \text{ in alternate segment})$$

$$\therefore 2\alpha + \beta = 180$$

(B)

$$3 \quad P(x) = (x^2 + 1)Q(x) + (2x + 3)$$

$$+ Q(x) = (x^2 + bx + c)$$

$$c + 1 = 0 \quad \therefore c = -1$$

$$b + 2 = a \quad b = 0$$

$$\therefore a = 2 \quad (B)$$

$$4 \quad \log_2 a$$

$$= (2^2)^{\log_2 a}$$

$$= 2^{\log_2 a^2}$$

$$= a^2 \quad (B)$$

$$5 \quad \sin^2 A - \sin A \cos A - \sin^2 A - \sin A \cos A$$

$$= \sin^2 A - \cos^2 A$$

$$= -2 \sin A \cos A$$

$$= -(\cos^2 A - \sin^2 A)$$

$$= \frac{-(\sin 2A)}{-(\cos 2A)}$$

$$= \tan 2A \quad (C)$$

$$6 \quad \frac{d}{dx} \sin^{-1}(2x-1)$$

$$\text{let } u = 2x-1$$

$$\frac{du}{dx} = 2$$

$$\frac{dy}{du} \times \frac{du}{dx}$$

$$= \frac{2}{\sqrt{1-(2x-1)^2}}$$

$$y = \sin^{-1} u$$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$$

$$= \frac{2}{\sqrt{1-4x^2+4x-1}}$$

$$= \frac{2}{\sqrt{4x(1-x)}}$$

$$= \frac{2}{2\sqrt{x(1-x)}}$$

$$= \frac{1}{\sqrt{x(1-x)}} \quad (D)$$

$$7 \quad \int \frac{1}{25(x^2 + \frac{9}{25})} dx$$

$$= \frac{1}{25} \times \frac{5}{3} \tan^{-1} \frac{5x}{3} + C$$

$$= \frac{1}{15} \tan^{-1} \frac{5x}{3} + C \quad (A)$$

$$8 \quad x = 2 \cos(3t + \frac{\pi}{6})$$

$$\text{max speed at } x = 0$$

$$\therefore 3t + \frac{\pi}{6} = \frac{\pi}{2}$$

$$3t = \frac{\pi}{3}$$

$$t = \frac{\pi}{9}$$

(B)

$$x = -6 \sin(3t + \frac{\pi}{6})$$

$$\text{speed} = |-6 \sin(\frac{\pi}{3} + \frac{\pi}{6})| = 6 \text{ m/s}$$

$$7 \quad |2x-1| \leq |x-2|$$

C.P

$x < \frac{1}{2}$	$\frac{1}{2} < x < 2$	$x > 2$
$2x+1 = -x+2$	$2x-1 = -x+2$	$2x-1 = x-2$
$-x = 1$	$3x = 3$	$x = -1$
$x = -1$	$x = 1$	X

$$\begin{array}{c} x \quad \checkmark \quad x \\ \leftarrow \quad -1 \quad 1 \quad \rightarrow \end{array}$$

$$x = -2 \quad 5 \leq 4 \quad \times$$

$$x = 0 \quad 1 \leq 2 \quad \checkmark$$

$$x = 2 \quad 3 \leq 0 \quad \times$$

$$-1 \leq x \leq 1 \quad \textcircled{C}$$

$$\frac{0}{\frac{dr}{dt}} = 0.02 \quad \frac{dA}{dt} = ?$$

$$A = \pi r^2$$

$$\frac{dA}{dr} = 2\pi r \quad \frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 2\pi r \times 0.02$$

$$\text{at } r=5$$

$$\frac{dA}{dt} = 2 \times \pi \times 5 \times 0.02$$

$$= \frac{\pi}{5} \text{ cm}^2/\text{sec}$$

$\textcircled{B}$

$$\frac{3x-2}{x+1} > 2$$

C.P at

$$x = -1$$

$$3x-2 = 2x+2$$

$$x = 4$$

$$\begin{array}{ccc} \checkmark & x & \checkmark \\ \leftarrow & -1 & 4 & \rightarrow \\ x = -2 & \frac{-8}{-1} > 2 & \checkmark \\ x = 0 & \frac{-2}{-1} > 2 & \times \\ x = 5 & \frac{13}{6} > 2 & \checkmark \end{array}$$

$$\frac{1}{x < -1}, \frac{1}{x > 4}$$

(2)

$$M(-1, 7) \sim (3, 1)$$

$$k: -1 \quad \frac{1}{(-5, 2)}$$

$$-5 = \frac{3k+1}{k-1}$$

$$-5k+5 = 3k+1$$

$$4 = 8k$$

$$k = \frac{1}{2}$$

(2)

$$\int \frac{1+6x}{1+x^2} dx$$

$$= \int \left(\frac{1}{1+x^2} + \frac{6x}{1+x^2} \right) dx$$

$$= \tan^{-1} x + 3 \ln(1+x^2) + c \quad (2)$$

$$1) y = x^3$$

$$y = 2-x^2$$

$$y' = 3x^2$$

$$y' = -2x$$

$$m_1 = 3$$

$$m_2 = -2$$

$$\tan \alpha = \left| \frac{3+2}{1-6} \right|$$

$$= |1|$$

$$\therefore \alpha = 45^\circ$$

(3)

e) Prove

$$1^2 + 3^2 + \dots + (2n-1)^2 = \frac{n}{3} (2n-1)(2n+1)$$

Prove $n=1$

$$LHS = 1^2$$

$$RHS = \frac{1}{3} (2-1)(2+1)$$

$$= 1$$

$$= 1$$

$$\therefore LHS = RHS$$

$$\therefore \text{true } n=1$$

1

Assume true $n=k$ k is the integer

ie

$$1^2 + 3^2 + \dots + (2k-1)^2 = \frac{k}{3} (2k-1)(2k+1)$$

Prove true $n=k+1$

$$LHS = 1^2 + 3^2 + \dots + (2k-1)^2 + (2(k+1)-1)^2$$

$$= \frac{k}{3} (2k-1)(2k+1) + (2k+2-1)^2$$

$$= \frac{k}{3} (2k-1)(2k+1) + (2k+1)^2$$

$$= (2k+1) \left[\frac{k}{3} (2k-1) + (2k+1) \right]$$

$$= (2k+1) \left(\frac{2k^2}{3} - \frac{k}{3} + 2k+1 \right)$$

$$= \frac{1}{3} (2k+1) (2k^2 - k + 6k + 3)$$

$$= \frac{1}{3} (2k+1) (2k^2 + 5k + 3)$$

$$= \frac{1}{3} (2k+1) (2k+3)(k+1)$$

$$= \frac{k+1}{3} (2(k+1)-1)(2(k+1)+1)$$

$\therefore$ true $n=k+1$

(3)

$$f) \int_0^2 \frac{x^2}{(x^3-1)^2} dx$$

$$u = x^3 - 1$$

$$\frac{du}{dx} = 3x^2$$

$$\frac{1}{3} \int_{-1}^7 \frac{1}{u^2} du$$

$$x=2 \quad u=7$$

$$x=0 \quad u=-1$$

$$= \frac{1}{3} \int_{-1}^7 u^{-2} du$$

$$= -\frac{1}{3} \left[u^{-1} \right]_{-1}^7$$

$$= -\frac{1}{3} \left(\frac{1}{7} + 1 \right)$$

$$= -\frac{8}{21}$$

(3)

Q12

$$a) \quad b-a = c-b = d$$

$$\frac{e^b}{e^a} = e^{b-a}$$

$$= e^{c-b} \quad (\text{from above})$$

$$= \frac{e^c}{e^b}$$

$$\therefore e^a, e^b, e^c \text{ are in G.P.} \quad (2)$$

$$b) \quad \angle ACB = 90^\circ \quad (\angle \text{ in semi-circle})$$

$$\angle BAC = \angle CAD$$

$$(\text{given } AC \text{ bisects } \angle DAB)$$

$$\angle DCA = \angle CBA \quad (\angle \text{ in alternate segment})$$

$$\therefore \triangle ABC \cong \triangle ACD \quad (\text{equiangular})$$

$$\therefore \angle ADC = \angle ACB = 90^\circ$$

$$(\text{corresponding } \angle \text{'s in similar } \triangle \text{'s}) \quad (3)$$

$$c) \quad P(x) = (x^2-1)Q(x) + ax+b$$

$$P(-1) = 0$$

$$P(1) = 2$$

$$\therefore -a+b=0$$

$$a+b=2$$

$$2b=2$$

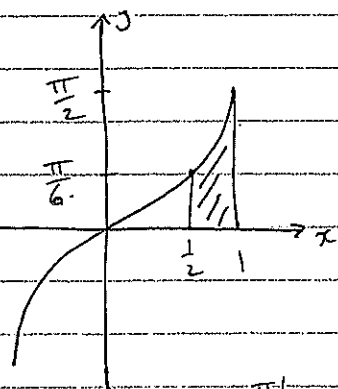
$$b=1$$

$$a=1$$

$$\therefore \text{Remainder} = x+1$$

(3)

1)



$$\text{Area} = \frac{\pi}{2} - \frac{\pi}{6} \times \frac{1}{2} - \int_{\pi/6}^{\pi/2} \sin y \, dy$$

$$= \frac{\pi}{2} - \frac{\pi}{12} + \left[\cos y \right]_{\pi/6}^{\pi/2} \leftarrow 1$$

$$= \frac{5\pi}{12} + \left(0 - \frac{\sqrt{3}}{2} \right) \quad 1$$

$$= \frac{5\pi - 6\sqrt{3}}{12} \quad \text{u2} \quad (3)$$

2) $y = \frac{2x^2 + 1}{x^2 - 4x}$

vert. asympt at

$$x(x-4) = 0$$

$$x=0, x=4 \quad 1$$

horiz asympt at

$$\lim_{x \rightarrow \infty} \frac{2x^2}{x^2} + \frac{1}{x^2} = \frac{x^2}{x^2} - \frac{4x}{x^2}$$

$$y = 2 \quad 1$$

(2)

e) $1 - \tan^2 x + \tan^4 x \dots$

limiting sum if $|r| < 1$

$$\therefore |- \tan^2 x| < 1 \quad 1$$

$$-1 < \tan x < 1$$

$$-\frac{\pi}{4} < x < \frac{\pi}{4} \quad \text{but } 0 \leq x < 2\pi$$

$$0 \leq x < \frac{\pi}{4} \quad \frac{3\pi}{4} < x < \frac{5\pi}{4} \quad \frac{7\pi}{4} < x < 2\pi$$

↑ any 1 (2)

13 $\frac{\pi}{3}$

a) LHS = $\int_{\pi/4}^{\pi/3} \sin^2 3x \, dx$

$$\cos 6x = 1 - 2\sin^2 3x$$

$$\sin^2 3x = \frac{1}{2} - \frac{1}{2} \cos 6x$$

$$= \frac{1}{2} \int_{\pi/4}^{\pi/3} (1 - \cos 6x) \, dx \quad 1$$

$$= \frac{1}{2} \left[x - \frac{\sin 6x}{6} \right]_{\pi/4}^{\pi/3} \quad 1$$

$$= \frac{1}{2} \left(\frac{\pi}{3} - \frac{\sin 2\pi}{6} - \frac{\pi}{4} + \frac{\sin \frac{3\pi}{2}}{6} \right)$$

$$= \frac{1}{2} \left(\frac{\pi}{3} - \frac{\pi}{4} - \frac{1}{6} \right)$$

$$= \frac{1}{2} \left(\frac{\pi - 2}{12} \right) \quad 1 \quad (3)$$

b) OQ: $m = \frac{aq^2}{2aq}$ OP: $m = \frac{p}{2}$

$$m_1 = \frac{q}{2}$$

$$m_1 m_2 = -1$$

$$\therefore \text{OQ} \perp \text{OP}$$

$$\therefore \frac{q}{2} \times \frac{p}{2} = -1$$

$$\therefore pq = -4 \quad 1$$

(1)

i) m is midpoint of OR

(diagonals of rectangle bisect each other)

$R(x, y)$

$$\therefore a(p+q) = \frac{x+0}{2}$$

(1)

$$\therefore x = 2a(p+q)$$

$$\frac{1}{2}a(p^2+q^2) = \frac{y+0}{2}$$

$$\therefore y = a(p^2+q^2)$$

$$\therefore R(2a(p+q), a(p^2+q^2))$$

i) $p+q = \frac{x}{2a}$

$$+ y = a((p+q)^2 - 2pq) \quad (1)$$

$$= a\left(\frac{x^2}{4a^2} - 2(-4)\right)$$

$$= a\left(\frac{x^2}{4a^2} + 8\right) \quad \checkmark$$

$$= \frac{x^2}{4a} + 8a$$

$$x^2 = 4a(y - 8a) \quad (2)$$

c) $\ddot{x} = 4(x+1)$

at $t=0, x=0, v=2$

i) $\frac{d}{ds}\left(\frac{1}{2}v^2\right) = 4x+4$

$$\frac{1}{2}v^2 = 2x^2 + 4x + c \quad \checkmark$$

at $x=0, v=2 \quad \therefore c=2$

$$v^2 = 4x^2 + 8x + 4$$

$$v^2 = 4(x^2 + 2x + 1)$$

$$= 4(x+1)^2$$

$$\therefore v = \sqrt{4(x+1)^2} \quad \checkmark$$

$$v = 2(x+1)$$

(2)

ii) $\therefore \frac{dx}{dt} = 2(x+1)$

$$\frac{dx}{2(x+1)} = \frac{1}{2} ds$$

$$\therefore t = \frac{1}{2} \int \frac{1}{x+1} ds$$

$$t = \frac{1}{2} \ln(x+1) + c \quad \checkmark$$

at $t=0, x=0$

$$\therefore 0 = \frac{1}{2} \ln 1 + c \quad \therefore c=0$$

$$\therefore t = \frac{1}{2} \ln(x+1)$$

$$2t = \ln(x+1)$$

$$e^{2t} = x+1$$

$$x = e^{2t} - 1 \quad \checkmark$$

(2)

d) i) $f(x) = e^x + x - 3$

$$f(0) = e^0 + 0 - 3$$

$$= 1 - 3$$

$$= -2 < 0 \quad \checkmark$$

$$f(1) = e + 1 - 3$$

$$= e - 2 > 0 \quad \checkmark$$

$$\therefore f(0) < 0, f(1) > 0$$

$\therefore$ root lies between

$$x=0 \text{ and } x=1$$

(2)

$$1) a = 0.8$$

$$f(x) = e^x + x - 3$$

$$f'(x) = e^x + 1$$

$$f(0.8) = e^{0.8} - 2.2$$

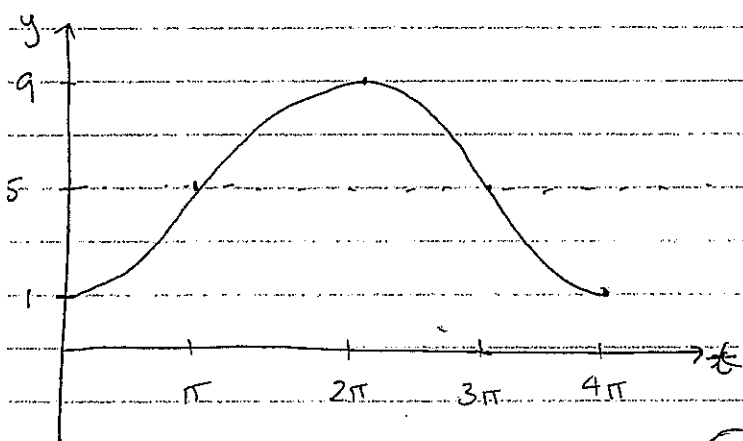
$$f'(0.8) = e^{0.8} + 1$$

$$a = 0.8 - \frac{e^{0.8} - 2.2}{e^{0.8} + 1}$$

$$a = 0.792 \text{ (3 s.f.)}$$

(2)

$$4) a) i) y = 5 - 4 \cos \frac{t}{2}$$



(2)

$$1) 3 = 5 - 4 \cos \frac{t}{2}$$

$$\cos \frac{t}{2} = \frac{1}{2}$$

$$\therefore \frac{t}{2} = \frac{\pi}{3}$$

$$t = \frac{2\pi}{3} h$$

$$= 2.09 h$$

$\therefore$ earliest time at 3:06 pm
(nearest minute)

(2)

$$b) N = 500 - 400e^{-0.1t}$$

$$\begin{aligned} i) \frac{dN}{dt} &= -400 \times 0.1 e^{-0.1t} \\ &= -0.1(-400e^{-0.1t}) \\ &= -0.1(500 - 400e^{-0.1t} - 500) \\ &= -0.1(N - 500) \end{aligned}$$

(1)

$$ii) \text{ at } t = 0$$

$$\frac{dN}{dt} = 40$$

$$\therefore \frac{dN}{dt} = 20$$

$$20 = -0.1(N - 500)$$

$$N = 300$$

(2)

$$c) i) 10! = 3628800$$

(1)

$$ii) 7! \times 4! = 120960$$

(2)

$$iii) 3628800 - 120960 = 3507840$$

(1)

$$d) i) (1.8 - h)^2 + x^2 = 2^2$$

$$(1.8 - h)^2 = 4 - x^2$$

$$1.8 - h = \sqrt{4 - x^2}$$

$$h = 1.8 - \sqrt{4 - x^2}$$

(1)

$$ii) \text{ at } x = 1.6 m$$

$$\frac{dh}{dx} = -\frac{1}{2}(4 - x^2)^{-1/2} \times -2x$$

$$= \frac{x}{\sqrt{4 - x^2}}$$

1

$$4 \frac{dh}{dt} = 0.5$$

$$\therefore \frac{dx}{dt} = \frac{dsc}{dh} \times \frac{dh}{dt} \quad \text{at } x = 1.6$$

$$\frac{dx}{dt} = \frac{\sqrt{4-x^2}}{x} \times 0.5 \quad \checkmark$$

$$= \frac{\sqrt{4-1.6^2}}{1.6} \times 0.5$$

$$= 0.375 \text{ m/s} \quad \checkmark$$

(3)